

Fedor Rodionov

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SUMMARY

PhD student in Generative AI at KAUST with deep expertise in Computer Vision, Geometry Processing, and Numerical Optimization. Strong R&D background in developing Neural Network systems, high-performance solvers, spatial algorithms, and data visualization applications. Proficient in C++ and Python, passionate about applying cutting-edge AI and graphics research to create next-generation visual experiences.

EDUCATION

- **King Abdullah University of Science and Technology (KAUST)** 2024 – Present
PhD in Computer Science (Generative AI); Advisor: Prof. Peter Wonka; GPA: 3.95/4.00
- **Lobachevsky State University of Nizhny Novgorod (UNN)** 2015 – 2021
B.Sc. (2019) and M.Sc. (2021) in Computer Science; GPA: 5.00/5.00
- **Yandex School of Data Analysis (YSDA)** 2019 – 2021
M.Sc.-equivalent program in Machine Learning (ML Developer Track); GPA: 4.83/5.00

EXPERIENCE

- **Huawei R&D - R&D Developer** Dec. 2022 - Aug. 2024
 - Developed and implemented advanced features for second-order numerical optimization solvers (FSQP, IpOpt) solving complex applied problems.
 - Researched state-of-the-art constrained non-convex optimization and second-order methods, enhancing solver capabilities with warm restart and LFBGS.
- **X5 Retail Group - Data Analyst** Feb. 2021 - Dec. 2022
 - Developed "A/B Helper" Python package to automate data collection, parameter selection, and prediction.
 - Built high-performance analytics dashboards using PySpark, Airflow, and ClickHouse for large-scale data.
- **Gorky.AI - Data Analyst** Jun. 2020 - Feb. 2021
 - Developed a GNN/NLP-based text clustering app; secured 2nd place in Moscow Mayor Hackathon.
 - Visualized complex geospatial data (urban infrastructure polygons) using Kepler.gl.
- **Yandex (Maps & Navigator) - Software Engineer Intern** Jan. 2020 - May. 2020
 - Optimized C++ and Boost.Python pipeline for road graph processing, making calculation time 6x faster.
 - Developed and implemented features and hotfixes for iOS and Android mobile applications, enhancing UX.

PUBLICATIONS

- **FloorplanQA: A Benchmark for Spatial Reasoning in LLMs using Structured Representations (2025):**
arXiv preprint, Under Review at ICML 2026 (arXiv:2507.07644)

TEACHING

- **KAUST - Team Assistant, Neural Network Design and Training** 2025
- **HSE University - Team Assistant, Python and Data Analysis** 2023-2024

PROJECTS

- **Deepworks:** Developed Deepworks, an open-source C++ framework for efficient neural network training and inference, demonstrating software engineering and deep learning expertise. (GitHub link)
- **3D City Map:** Created 3D city map visualization of building data using Kepler.gl and GeoPandas, showcasing computer graphics and geometry processing skills. (GitHub link)

TECHNICAL SKILLS

- **Languages:** C++, Python, SQL, English (IELTS 7.5)
- **Technologies:** PyTorch, HuggingFace, ComfyUI, PySpark, AWS, Git, Docker, Airflow, Blender
- **Core:** Generative AI, Numerical Optimization, Deep Learning, Computer Vision, Image Diffusion